

Throughput and Acceleration Trade-offs of Lightweight Block Ciphers: A CPU and GPU Comparison of PRESENT-80, SPECK-64/128, and AES-128

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Abstract—Lightweight block ciphers are designed to fit within the stringent silicon area budgets of constrained IoT hardware where standards like AES-128 are too heavy. However, the software and GPU throughput characteristics of these architectural families are rarely compared against AES on a single, controlled platform. This paper presents from-scratch C++ and CUDA implementations of PRESENT-80, SPECK-64/128, and AES-128, validated against official test vectors and benchmarked on an AMD EPYC CPU and an NVIDIA L40S GPU. Our evaluation shows that the Add-Rotate-XOR (ARX) design of SPECK achieves massive performance advantages, leading CPU and GPU encryption throughput at 1,133 MB/s and 356 GB/s, respectively, while minimizing key-schedule latency and device memory footprint. Conversely, the Substitution-Permutation Network (SPN) design of PRESENT-80 yields the lowest absolute throughput but achieves the highest relative GPU speedup (over 5,200 $\times$) by mitigating severe CPU-side bit-manipulation inefficiencies. Furthermore, we identify a 37% performance drop specific to AES-128 in Counter (CTR) mode on the GPU, driven by per-thread register pressure inherent to its table-intensive structure. These insights demonstrate that ARX ciphers uniquely align hardware area constraints with software and parallel processing scalability, providing critical design guidance for secure IoT ecosystems.

Index Terms—lightweight cryptography, block cipher, PRESENT, SPECK, AES, GPU, CUDA, IoT security, throughput

I. INTRODUCTION

The proliferation of IoT devices has created demand for encryption that operates within tight resource budgets. The smallest constrained targets, including passive RFID tags, medical implants, and sensor nodes, impose silicon area budgets near 2,000 gate equivalents (GE), memory as small as a few hundred bytes, and power budgets measured in microwatts [7]. AES-128 requires approximately 3,400 GE in hardware [3], which exceeds the IoT budget and motivates dedicated lightweight cipher design.

Two prominent design families have emerged. Substitution-Permutation Network (SPN) ciphers, exemplified by

PRESENT-80 [1], use structured S-box substitution and bit-permutation rounds. Their permutation layers are wire-routing in dedicated silicon and cost essentially nothing in hardware area, but become explicit bit-manipulation operations in software. Add-Rotate-XOR (ARX) ciphers, exemplified by SPECK-64/128 [2], use only modular addition, rotation, and XOR per round. These operations map directly to native CPU and GPU integer arithmetic units with no table lookups, making ARX designs fast in software while remaining compact in hardware.

Prior work reports hardware GE footprints for these ciphers in isolation. A direct platform-controlled comparison of CPU throughput, GPU throughput, key-schedule cost, and mode behavior on modern server hardware has not been published for these specific cipher families. This paper fills that gap with a consistent implementation and measurement methodology.

The contributions of this work are as follows. We implement all three ciphers from scratch in C++ and CUDA without any third-party cryptographic libraries, hardware intrinsics, or vendor-optimized paths, and validate each against its official reference vectors. We measure ECB, CTR, and CBC CPU throughput across five buffer sizes spanning 1 KB to 64 MB. We characterize GPU ECB and CTR throughput on an NVIDIA L40S and derive speedup ratios relative to the CPU baseline. We measure key-schedule latency and device memory footprint per cipher. We analyze the resulting trade-off between hardware area, software throughput, GPU scalability, and standardization coverage.

The remainder of this paper is organized as follows. Section II provides background on cipher construction and hardware footprint. Section III describes the implementation. Section IV describes the experimental setup. Section V presents results. Section VI discusses the implications. Section VII concludes.

TABLE I
CIPHER SPECIFICATIONS

Cipher	Type	Block	Key	Rounds
PRESENT-80	SPN	64-bit	80-bit	31
SPECK-64/128	ARX	64-bit	128-bit	27
AES-128	SPN	128-bit	128-bit	10

TABLE II
PUBLISHED HARDWARE AREA ESTIMATES

Cipher	Area (GE)	Fits <2,000 GE
PRESENT-80	$\approx 1,570$ [1]	Yes
SPECK-64/128	$\approx 1,280$ [2]	Yes
AES-128	$\approx 3,400$ [3]	No

II. BACKGROUND

A. Cipher Constructions

An SPN cipher alternates substitution and permutation layers in each round. Substitution provides nonlinearity through S-box lookups and permutation diffuses S-box outputs across the full state. PRESENT-80 uses a 4-bit S-box and a deterministic bit permutation that maps bit position j to position $(16j) \bmod 63$. AES-128 uses an 8-bit S-box, ShiftRows, MixColumns over $\text{GF}(2^8)$, and AddRoundKey. The AES S-box and its inverse together occupy 512 bytes of device constant memory in the GPU implementation.

An ARX cipher applies one round of modular addition, word rotation, and XOR. SPECK-64/128 computes $x \leftarrow (\text{ROR}(x, 8) + y) \oplus k$ and $y \leftarrow \text{ROL}(y, 3) \oplus x$ per round. This requires no lookup tables, so SPECK’s GPU device footprint consists entirely of round keys.

Table I summarizes the cipher parameters.

B. Hardware Area Footprint

Table II lists published hardware area estimates for each cipher. SPECK-64/128 and PRESENT-80 both fit within the 2,000 GE IoT area budget. AES-128 does not.

C. Modes of Operation

ECB (Electronic Codebook) encrypts each block independently and serves as the primary throughput benchmark mode in this work [5]. CTR (Counter mode) encrypts successive counter values and XORs each result with the corresponding plaintext block, making all blocks fully independent and suitable for parallel GPU execution. CBC (Cipher-Block Chaining) XORs each plaintext block with the previous ciphertext before encryption, making forward encryption sequential while allowing parallel decryption.

III. IMPLEMENTATION

A. CPU Implementation

All three ciphers are implemented in C++17 and compiled with `g++ 13.3 -O3`. The host interface follows a uniform batch API in which a caller provides a key, plaintext buffer, and block count. PRESENT-80 and SPECK-64/128 share this

TABLE III
KEY SCHEDULE AND DEVICE CONSTANT MEMORY FOOTPRINT

Cipher	RK (B)	S-box (B)	Total (B)	KS (μs)	KS cyc
PRESENT-80	256	32	288	2.20	8,465
SPECK-64/128	108	0	108	0.026	100
AES-128	176	512	688	0.094	360

interface and are validated against the official CHES 2007 and NSA 2013 reference vectors, respectively. AES-128 is validated against FIPS-197 Appendix B and Appendix C test vectors. SPECK-64/128 decryption was explicitly optimized for CPU to reduce per-block cycle count relative to a direct inverse of the encryption schedule.

B. GPU Implementation

All CUDA kernels are compiled with `nvcc -O3` targeting CUDA 12.8 [6]. Each kernel launches one thread per cipher block, which allows the GPU scheduler to assign independent AES, SPECK, or PRESENT rounds to threads without any synchronization.

Round keys are computed entirely on the CPU host and uploaded to device constant memory via `cudaMemcpyToSymbol`. Constant memory is a read-only GPU address space that is cached and broadcast to all threads in a warp at zero per-thread cost. Key-schedule time is excluded from all GPU kernel measurements because a single schedule is amortized over all blocks processed under one key. ECB and CTR kernels are provided for all three ciphers. The CTR kernel constructs an independent counter per thread from the block index and XORs the encrypted counter with the corresponding plaintext block.

Table III lists the device constant memory breakdown per cipher. SPECK-64/128 requires no S-box and occupies only 108 bytes. AES-128 stores forward and inverse S-boxes (256 bytes each) plus round keys (176 bytes) for a total of 688 bytes in ECB mode, and 432 bytes in CTR mode because CTR requires only the forward S-box. PRESENT-80 stores 256 bytes of round keys and 32 bytes for its compact 4-bit S-box pair (288 bytes total for ECB, 272 bytes for CTR with only the forward S-box).

C. Correctness Validation

All six mode roundtrip tests (ECB, CTR, and CBC for each cipher) across the CPU implementations pass complete encrypt-decrypt verification. For the GPU, CBC mode was not implemented due to its inherent block-to-block data dependencies, which prevent parallelization and conflict with the GPU’s multi-threaded execution model. No timing data is recorded unless the corresponding validation check passes.

IV. EXPERIMENTAL SETUP

All experiments run on an AMD EPYC 9374F 32-core processor paired with an NVIDIA L40S GPU [8]. Each measurement is the mean of five independent trials. Buffer

sizes sweep from 1 KB to 64 MB in five steps (1 KB, 64 KB, 1 MB, 16 MB, 64 MB), spanning a 65,536-fold range of input volume to characterize small-buffer overhead and large-buffer steady-state throughput separately.

GPU measurements record kernel execution time only, excluding host-to-device data transfer. CPU measurements use the RDTSC hardware timestamp counter for cycle-accurate timing. All benchmarks run in ECB mode unless otherwise noted. Mode-overhead measurements (CTR and CBC) use fixed 1 MB payloads for the CPU and the full sweep for GPU CTR.

V. RESULTS

A. CPU Throughput

Table IV shows CPU ECB encryption and decryption throughput. SPECK-64/128 sustains approximately 1,133 MB/s for encryption at all buffer sizes, which is 47 times faster than AES-128 at 23.87 MB/s and 784 times faster than PRESENT-80 at 1.4451 MB/s at 64 MB. Throughput is nearly flat across buffer size for all three ciphers, indicating that each workload is compute-bound with the cipher state resident in processor cache.

AES-128 decryption is considerably slower than encryption at approximately 15.0 MB/s across all buffer sizes, because the inverse cipher path requires additional table accesses that have no hardware equivalent at this optimization level. PRESENT-80 is nearly symmetric, reaching 1.4270 MB/s in decryption at 64 MB. SPECK-64/128 decryption is somewhat slower than encryption, at 850.54 MB/s versus 1,132.77 MB/s at 64 MB, reflecting the sequential inverse key-schedule traversal required per block.

The per-block cost explains the throughput ordering directly. SPECK-64/128 requires 39 cycles per block and 3.10 cycles per byte. AES-128 requires 2,580 cycles per block and 161 cycles per byte. PRESENT-80 requires 21,136 cycles per block and 734 cycles per byte, as shown in Table V.

B. GPU ECB Throughput and Speedup

At 64 MB, SPECK-64/128 reaches 356,150 MB/s in ECB encryption, AES-128 reaches 32,672 MB/s, and PRESENT-80 reaches 7,519 MB/s. At 1 KB, all three ciphers underperform (SPECK 571.7, AES 88.9, PRESENT 43.9 MB/s) because the buffer holds too few blocks to saturate the L40S. Throughput rises sharply between 64 KB and 1 MB as GPU parallelism fills the device, and stabilizes above 16 MB.

SPECK-64/128 shows an elevated peak at 16 MB of 823,317 MB/s in ECB encryption, 2.3 times its 64 MB value. This behavior is discussed in Section VI.

GPU speedup over the CPU ECB baseline at 64 MB is approximately 5,203 times for PRESENT-80, 1,370 times for AES-128, and 314 times for SPECK-64/128. The speedup ordering inverts the absolute throughput ordering: the cipher with the weakest CPU performance (PRESENT-80) gains the largest relative multiplier. Fig. 1 shows the full throughput and speedup scaling curves for SPECK-64/128. Fig. 2 and Fig. 3

show the corresponding curves for PRESENT-80 and AES-128.

C. GPU CTR Mode Performance

Table VI reports GPU peak throughput at 64 MB for ECB and CTR modes. PRESENT-80 and SPECK-64/128 show negligible difference between modes. SPECK-64/128 ECB encryption is 356,150 MB/s and CTR is 357,693 MB/s, a difference below 0.5 percent. PRESENT-80 shows 7,519 versus 7,502 MB/s, within 0.2 percent.

AES-128 behaves differently. Its CTR encryption is 20,626 MB/s at 64 MB, which is 37 percent below its ECB value of 32,672 MB/s. The same gap appears at 1 MB (CTR 17,536 versus ECB 26,580 MB/s) and 16 MB (CTR 20,554 versus ECB 32,461 MB/s). The cause is analyzed in Section VI.

D. Key Schedule Analysis

PRESENT-80 is the most expensive cipher to initialize at $2.20 \mu\text{s}$ and 8,465 cycles, approximately 85 times slower than SPECK-64/128 (100 cycles) and 23.5 times slower than AES-128 (360 cycles). The high PRESENT-80 cost reflects its 31-round key extraction from an 80-bit key register. SPECK-64/128 has the smallest device footprint at 108 bytes, compared to 288 bytes for PRESENT-80 and 688 bytes for AES-128 in ECB mode. Notably, the AES-128 S-box tables alone (512 bytes) exceed the entire SPECK-64/128 device footprint. See Table III in Section III.

E. Mode Overhead on CPU

Table VII shows CPU throughput for ECB, CTR, and CBC. AES-128 changes little across modes: ECB 23.6, CTR 21.56, CBC 22.08 MB/s. The cipher is slow enough that per-block mode bookkeeping is a negligible fraction of total work. SPECK-64/128 shows the largest relative penalty: CTR drops 53 percent from ECB (528.7 versus 1,133 MB/s) and CBC drops 74 percent (291.0 MB/s). A fast cipher makes per-block counter increments and sequential CBC chaining costs visible. PRESENT-80 shows little change across modes in relative terms (ECB 1.45, CTR 1.40, CBC 1.33 MB/s) because round computation dominates all other overhead.

VI. DISCUSSION

Largest speedup versus lowest absolute throughput. PRESENT-80 achieves a 5,203-times GPU speedup despite sustaining only 1.445 MB/s on the CPU, the largest relative gain in this study. The GPU hides per-thread latency by executing thousands of independent blocks concurrently, so the relative gain is largest precisely where the CPU baseline is weakest. This is a symptom of CPU inefficiency, not of intrinsic GPU friendliness. In absolute terms, PRESENT-80 still trails SPECK-64/128 on the GPU by approximately 47 times (7,519 versus 356,150 MB/s). A high speedup ratio alone does not indicate high absolute performance.

ARX aligns area and throughput. SPECK-64/128 achieves the smallest hardware footprint ($\approx 1,280$ GE), the

TABLE IV
CPU ECB THROUGHPUT (MB/s): ENCRYPTION AND DECRYPTION

Size	PRESENT enc	PRESENT dec	SPECK enc	SPECK dec	AES enc	AES dec
1 KB	1.4791	1.4662	1167.73	880.24	23.80	15.02
64 KB	1.4448	1.4255	1225.82	899.78	23.88	15.04
1 MB	1.4456	1.4282	1098.32	849.03	23.85	15.04
16 MB	1.4453	1.4272	1131.34	844.62	23.84	14.99
64 MB	1.4451	1.4270	1132.77	850.54	23.87	15.02

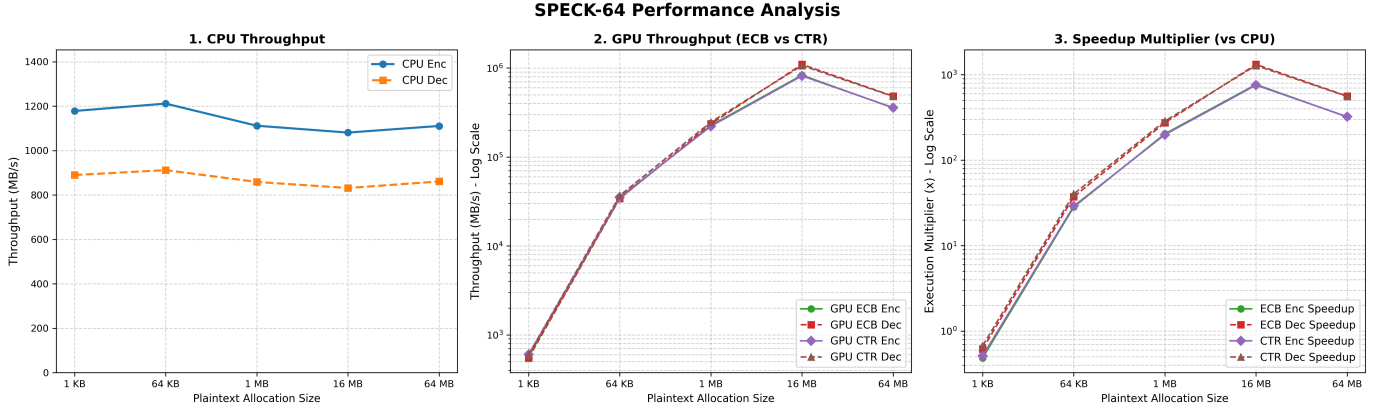


Fig. 1. SPECK-64/128 performance. Panel 1: CPU ECB throughput (linear scale). Panel 2: GPU ECB and CTR throughput (log scale). Panel 3: GPU speedup over CPU baseline (log scale).

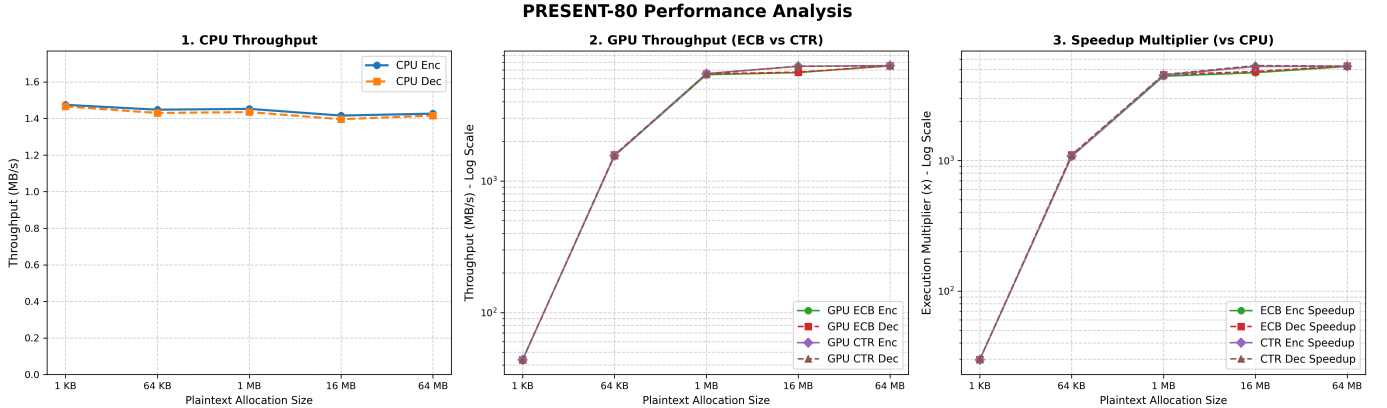


Fig. 2. PRESENT-80 performance. Panel 1: CPU ECB throughput (linear scale). Panel 2: GPU ECB and CTR throughput (log scale). Panel 3: GPU speedup over CPU baseline (log scale).

TABLE V
SINGLE-BLOCK CPU COST

Cipher	cyc/byte	Block cycles	Latency (μ s)
PRESENT-80	734	21,136	≈ 5.0
SPECK-64/128	3.10	39	≈ 0.009
AES-128	161	2,580	≈ 0.62

smallest device constant memory (108 bytes), and the highest CPU and GPU throughput among the three ciphers. These properties are not coincidental. ARX operations map directly to native integer ALUs with no table lookups and no ded-

icated S-box combinational logic. The same absence of S-box hardware that keeps silicon area small also eliminates memory bandwidth and instruction overhead in software. SPN ciphers do not share this alignment. The bit permutation that is cost-free in hardware (a set of wires) becomes explicit bit-manipulation in software, and the S-box that is compact in silicon requires memory access in software. SPECK demonstrates that a cipher can jointly minimize area and maximize throughput when the design avoids table-based operations entirely.

AES CTR throughput drop on GPU. The 37-percent gap between AES-128 ECB and CTR GPU throughput (32,672

TABLE VI
GPU PEAK THROUGHPUT AT 64 MB (MB/s): ECB AND CTR, ENCRYPTION AND DECRYPTION

Cipher	ECB enc	ECB dec	CTR enc	CTR dec	Speedup vs CPU (ECB enc)
PRESENT-80	7,518.84	7,509.61	7,501.53	7,502.71	$\approx 5,203\times$
SPECK-64/128	356,149.73	486,307.40	357,692.65	482,192.59	$\approx 314\times$
AES-128	32,671.82	35,436.36	20,626.12	20,638.33	$\approx 1,370\times$

AES-128 Performance Breakdown

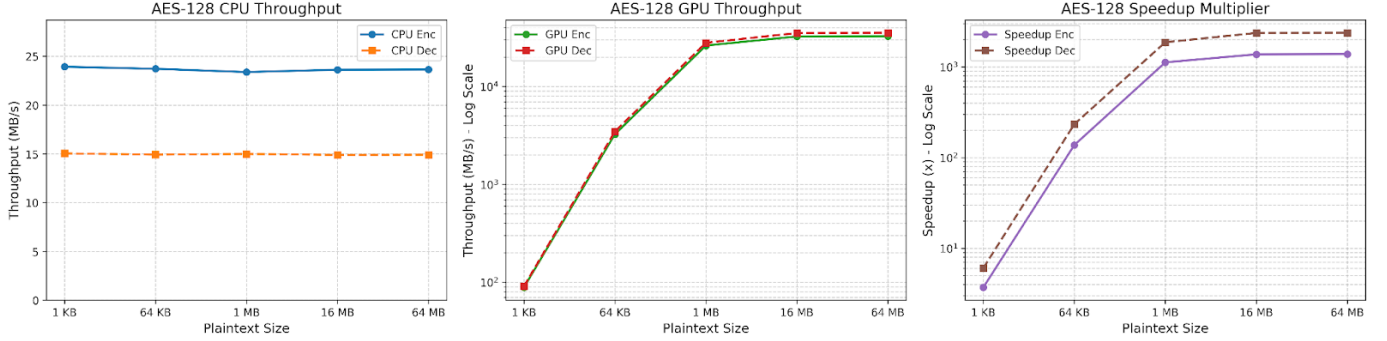


Fig. 3. AES-128 performance. Panel 1: CPU ECB throughput (linear scale). Panel 2: GPU ECB and CTR throughput (log scale). Panel 3: GPU speedup over CPU baseline (log scale).

TABLE VII
CPU THROUGHPUT BY MODE OF OPERATION (MB/s)

Cipher	ECB	CTR	CBC
PRESENT-80	≈ 1.45	1.40	1.33
SPECK-64/128	$\approx 1,133$	528.7	291.0
AES-128	≈ 23.6	21.56	22.08

versus 20,626 MB/s at 64 MB) does not appear in PRESENT-80 or SPECK-64/128, and warrants explanation. The AES CTR kernel uses only the forward S-box and requires 432 bytes of constant memory, less than the 688 bytes used by the ECB kernel. Lower constant memory usage should, in isolation, improve cache efficiency. The observed degradation therefore points to per-thread register pressure rather than constant cache capacity as the root cause. Constructing a 128-bit counter and feeding it through ten AES rounds requires more registers per thread than the direct plaintext path in ECB mode, reducing warp occupancy on the L40S. SPECK-64/128 and PRESENT-80 have sufficient register headroom for CTR’s counter construction at their respective round counts (27 and 31 rounds) and show no occupancy reduction. This finding suggests that the CTR mode penalty for AES is specific to table-intensive SPN kernels with high per-thread state requirements.

SPECK-64/128 throughput at 16 MB versus 64 MB. SPECK-64/128 ECB encryption peaks at 823,317 MB/s at 16 MB then settles to 356,150 MB/s at 64 MB. At 16 MB the working set interacts favorably with the L40S GPU L2 cache and the kernel is compute-bound throughout execution.

At 64 MB the buffer exceeds effective L2 capacity, the kernel transitions to a memory-bandwidth-limited regime, and throughput is bounded by the device memory bus rather than by the cipher’s arithmetic rate. We report the 64 MB value as the stable steady-state throughput for SPECK-64/128.

System design implications. For constrained devices where AES compliance is not mandated, SPECK-64/128 is the strongest choice across all software and GPU metrics, fitting within the 2,000 GE area budget while leading in throughput. PRESENT-80 is appropriate when ISO/IEC 29192-2 compliance [4] or a purely hardware-oriented SPN structure is required, with the understanding that its software throughput is poor by design. AES-128 is correct when FIPS-197 compliance is mandatory. Its CPU performance of 23.87 MB/s in this study uses no AES-NI instructions, so hardware-accelerated deployments would substantially exceed this baseline. The central conclusion is that hardware area, software throughput, GPU scalability, and cryptographic standardization rarely optimize together, and ARX ciphers such as SPECK come closest to satisfying the first three simultaneously.

VII. CONCLUSION

This paper presents from-scratch C++ and CUDA implementations of PRESENT-80, SPECK-64/128, and AES-128, validated against official test vectors and benchmarked on an AMD EPYC 9374F CPU and NVIDIA L40S GPU. SPECK-64/128 leads all three ciphers in CPU encryption throughput (1,133 MB/s), GPU peak throughput (356,150 MB/s), key-schedule speed (100 cycles), and device footprint (108 bytes). PRESENT-80 achieves the largest GPU speedup at 5,203 times over its CPU baseline but the lowest absolute throughput in

every configuration. AES-128 delivers the broadest standardization coverage and requires hardware AES instructions to be competitive on CPU throughput. Counter mode GPU throughput matches ECB for PRESENT-80 and SPECK-64/128 but falls 37 percent for AES-128, identifying a register-pressure effect tied to the AES table-intensive SPN kernel. Future work includes incorporating AES-NI for a fair modern comparison, porting to microcontroller-class targets, measuring end-to-end GPU latency including host-device transfer, and evaluating ASCON as the NIST Lightweight Cryptography competition winner.

SOURCE CODE

The full source code and benchmark logs are publicly available at

[https://github.com/devansh-gupta19/
ECE268-Lightweight_block_cipher](https://github.com/devansh-gupta19/ECE268-Lightweight_block_cipher)

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